

Security Event Analysis Report

1. Types of Security Logs Analyzed

The following three types of security logs were analyzed:

Log Type	Source	Description
Authentication Logs	Windows & Linux hosts	User login attempts (success and failure)
Cloud Authentication Logs	Microsoft 365 (Elastic SIEM)	Portal login activities from unusual locations
Malware Detection Logs	Microsoft 365 SharePoint (Elastic SIEM)	File uploads flagged as malicious

2. Interpretation Methodology

Each log type was parsed and interpreted using the following methodology:

- Timestamp Analysis:** Events were organized chronologically to detect suspicious sequences.
- Outcome Evaluation:** Focused on failed vs successful login attempts, particularly anomalous outcomes.
- Geolocation Inspection:** IP addresses were mapped to countries to detect abnormal access attempts.
- Cross-Referencing Fields:** `user.name`, `source.ip`, and `geo.country_name` fields were used to establish context and relationships.
- Correlation Between Sources:** Attempted to match suspicious login behavior with other alerts (malware detections, rare locations).

3. Correlation Between Log Sources (Demonstrated)

We demonstrated correlation between:

- Authentication Failures from Linux** (`/var/log/auth.log`)
- Rare Location Logins from Microsoft 365**

Example Correlation:

- **Failed SSH login** attempts were seen from IPs geolocated in North Korea and Iran (Linux syslog data).
- **Simultaneously, successful** logins into Microsoft 365 accounts from rare locations like Russia and Germany occurred (Microsoft 365 portal logs).

Finding:

- A failed Linux login from North Korea was closely followed by a successful Microsoft 365 portal login from Germany for user `cgonzalez`.
- Possible use of compromised credentials across platforms.

4. Complete Incident Detection Scenario

Scenario: Suspicious Cross-Platform Login Activity

Timeline:

Time (UTC)	Event
2025-04-19 17:04:58	Failed Windows login (User: <code>christopherking</code> , United States)
2025-04-19 23:39:43	Failed SSH login (User: <code>guest</code> , IP from North Korea)
2025-04-20 09:51:16	Failed SSH login (User: <code>demo</code> , IP from North Korea)
2025-04-25 10:30:02	Successful Microsoft 365 login (User: <code>cgonzalez</code> , Germany)

Correlations Observed:

- Unusual geographic pattern (North Korea and Germany, not typical locations).
- Failed attempts on Linux followed by a successful cloud login suggest stolen credentials used remotely.
- **Risk of credential compromise via brute force or phishing.**

Annotated Log Excerpt:

json
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```
{  
  "@timestamp": "2025-04-19T23:39:43",
```

```
"event.category": "authentication",
"event.action": "user_login",
"event.outcome": "failure",
"user.name": "guest",
"source.ip": "4.249.104.148",
"geo.country_name": "North Korea"
}
json
CopyEdit
{
  "@timestamp": "2025-04-25T10:30:02",
  "event.category": "authentication",
  "event.action": "user_login",
  "event.outcome": "success",
  "user.name": "cgonzalez",
  "source.ip": "112.233.155.75",
  "geo.country_name": "Germany"
}
```

5. Alert Triage Process

Severity Assessment:

- Successful login from rare locations → **Medium Severity** (per rule definition)
- Multiple failed Linux login attempts → **High Severity** if same usernames and foreign IPs involved

False Positive Identification:

- **Possible false positives** in cases where:
 - Legitimate users travel internationally.
 - VPNs or corporate proxies mask true location.

Escalation Criteria:

- Escalate if:
 - Multiple failed authentications immediately precede a successful login.
 - Access originates from blacklisted or high-risk countries (North Korea, Iran, Russia).

- The user has access to sensitive or administrative resources.

Example Escalation Decision:

- The chain involving user `cgonzalez` warranted escalation because:
 - Unusual successful login country (Germany)
 - Preceded by brute-force style failures
 - No travel evidence (mock data assumption)

6. Investigation Methodology Summary

Step	Activity
1	Collected authentication logs from Linux, Windows, and Microsoft 365
2	Parsed fields: timestamp, event.outcome, user.name, source.ip
3	Geolocated IPs for abnormal country detection
4	Correlated events across platforms using user accounts
5	Determined if behavior matched threat patterns (TA0001 Initial Access, per MITRE ATT&CK™)
6	Assessed severity, ruled out false positives, and documented conclusions

Final Conclusion

The security event analysis identified a **potential multi-stage intrusion attempt**, beginning with brute-force login failures on Linux and culminating in a suspicious Microsoft 365 portal access from a rare location. Based on the event correlation, timing, and source geolocation, the incident was escalated for further SOC review